Generating Functions.

<u>Definition</u>: Let $a_0, a_1, a_2, \ldots$ be a sequence of real numbers. The function $f(z) = a_0 + a_1 z + a_2 z^2 + \cdots$

$$= \sum_{i=0}^{\infty} a_i z^i$$

is called the generating function for the given sequence $\{a_i\}_{i \geq 0}$. The function $f(z)$ is a Taylor series in z and is convergent in some region $|z| < R$, where R is called the radius of convergence of the power series of $f(z)$. Each sequence has a unique generating function [redacted] $f(z)$ which can be expanded in a power series of z, and the coefficient of z^n gives the nth element a_n of the sequence. For example :-

<u>ex1.</u> $$(1+z)^n = \binom{n}{0} + \binom{n}{1} z + \binom{n}{2} z^2 + \cdots + \binom{n}{n} z^n,$$

so $(1+z)^n$ is a generating function for the sequence $\left\{ \binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \ldots, \binom{n}{n}, 0, 0, 0, \ldots \right\}$. This sequence is convergent in the region $|z| < \infty$, i.e., $R = \infty$.

<u>ex2.</u> $$(1 - z^{n+1}) = (1-z)(1 + z + z^2 + \cdots + z^n)$$
$$\text{for } n \in \mathbb{Z}^+$$

Implies, $$\frac{1 - z^{n+1}}{1 - z} = 1 + z + z^2 + \cdots + z^n,$$

and thus $\frac{1-z^{n+1}}{1-z}$ is the generating function for the sequence $\{1, 1, \cdots, 1, 0, 0, \cdots\}$ where first $(n+1)$ terms are 1, and the rests are zero. The radius of convergence of this generating function $f(z) = \frac{1-z^{n+1}}{1-z}$ is

$|z| < 1$, i.e., $R = 1$.

★ Note down following important generating functions:-

i) $1 = (1-z)(1+z+z^2+z^3+\cdots\cdots)$

$\Rightarrow \frac{1}{1-z} = 1 + z + z^2 + z^3 + \cdots\cdots\cdots\cdots$ —— (*)

Therefore, $f(z) = \frac{1}{1-z}$ is the generating function of the sequence $\{1, 1, 1, \cdots\cdots\}$. The radius of convergence of $f(z) = \frac{1}{1-z}$ is $|z| < 1$, i.e., $R = 1$.

ii) Differentiating the identity (*) with respect to 'z', one can obtain

$$\frac{1}{(1-z)^2} = 1 + 2z + 3z^2 + 4z^3 + \cdots\cdots$$

Thus, $f(z) = \frac{1}{(1-z)^2}$ is the generating function of the sequence $\{1, 2, 3, 4, \cdots\cdots\}$,

while $f(z) = \frac{z}{(1-z)^2} = 0 + 1.z + 2.z^2 + 3.z^3 + \cdots$ (**)

is the generating function of the sequence $\{0, 1, 2, 3, 4, \cdots\}$.

For the above generating functions, the radius of convergence $R = 1$, i.e., $f(z)$ is convergent in the region $|z| < 1$

(iii) Differentiating the above identity (**) with respect to 'z', one can obtain

$$\frac{z+1}{(1-z)^3} = 1 + 2^2 z + 3^2 z^2 + 4^2 z^3 + \cdots$$

Also, $$\frac{z(z+1)}{(1-z)^3} = z + 2^2 z^2 + 3^2 z^3 + 4^2 z^4 + \cdots$$

Therefore, $\frac{z+1}{(1-z)^3}$ generates the sequence $\{1^2, 2^2, 3^2, \cdots\}$ and $\frac{z(z+1)}{(1-z)^3}$ generates $\{0^2, 1^2, 2^2, 3^2, \cdots\}$, and both the generating functions are convergent in the region $|z| < 1$, i.e., radius of convergence $R = 1$

(iv) Using the above identity (*), one may write

$$\frac{1}{1-2z} = 1 + 2z + (2z)^2 + (2z)^3 + \cdots$$

Therefore $f(z) = \frac{1}{1-2z}$ is the generating function of the sequence $\{1 = 2^0, 2, 2^2, \cdots\}$. The radius of convergence for this generating function $f(z) = \frac{1}{1-2z}$ is given by $|z| < \frac{1}{2}$, i.e., $R = \frac{1}{2}$

Similarly, $\frac{1}{1-az} = 1 + az + a^2z^2 + a^3z^3 + \cdots\cdots,$

and one may note that $f(z) = \frac{1}{1-az}$ is the generating function of the sequence $\{1 = a^0, a, a^2, \cdots\cdots\}$, and $f(z) = \frac{1}{1-az}$ is convergent in the region $|z| < \frac{1}{a}$, i.e., the radius of convergence for $f(z) = \frac{1}{1-az}$ is given by $R = \frac{1}{a}$.

(v) Let $f(z) = \frac{1}{1-z} = 1 + z + z^2 + z^3 + \cdots\cdots$

~~Thus~~ $f(z) = \frac{1}{1-z}$ is the generating function of the sequence $\{1, 1, 1, 1, \cdots\cdots\}$.

Let $g(z) = f(z) - z^2 = \frac{1}{1-z} - z^2$.

Therefore $g(z)$ is the generating function of the sequence $\{1, 1, 0, 1, \cdots\cdots\}$, and

also $h(z) = f(z) + 2z^3$ is the generating function of the sequence

$$\{1, 1, 1, 3, 1, \cdots\cdots\}$$

The radius of convergence for $g(z)$, $f(z)$, and $h(z)$ are given by $R = 1$, i.e., $g(z)$, $f(z)$ and $h(z)$ converges in the region $|z| < 1$.

★ For $n \in \mathbb{Z}^+$, the binomial theorem tells us that $(1+z)^n = \binom{n}{0} + \binom{n}{1} z + \binom{n}{2} z^2 + \cdots + \binom{n}{n} z^n$

This idea may be extended when $n<0$ or n is not necessarily an integer.

Assume $n, r \in \mathbb{Z}^+$ and $n \geq r \geq 0$, we have

$$\binom{n}{r} = \frac{\lfloor n}{\lfloor r \, \lfloor n-r} = \frac{n(n-1)(n-2)(n-3)\cdots(n-r+1)}{\lfloor r}$$

If $n \in \mathbb{R}$, we use

$$\binom{n}{r} = \frac{n(n-1)(n-2)\cdots(n-r+1)}{\lfloor r}$$

For example, $n \in \mathbb{Z}^+$, we have

$$\binom{-n}{r} = \frac{(-n)(-n-1)(-n-2)\cdots(-n-r+1)}{\lfloor r}$$

$$= (-1)^r \binom{n+r-1}{r},$$

also for each real n, $\binom{n}{0} = 1$

One may give a formal proof of this result using Maclaurin series expansion of $(1+x)^{-n}$, where $n \in \mathbb{Z}^+$, is given by

$$(1+x)^{-n} = 1 + (-n)x + (-n)(-n-1)\frac{x^2}{\lfloor 2} + (-n)(-n-1)(-n-2)\frac{x^3}{\lfloor 3} + \cdots$$

$$\Rightarrow (1+x)^{-n} = 1 + \sum_{r=1}^{\infty} \frac{(-n)(-n-1)\cdots(-n-r+1)}{\lfloor r} x^r$$

$$= \sum_{r=0}^{\infty} (-1)^r \binom{n+r-1}{r} x^r$$

Hence $(1+x)^{-n} = \binom{-n}{0} + \binom{-n}{1}x + \binom{-n}{2}x^2 + \cdots = \sum_{r=0}^{\infty} \binom{-n}{r} x^r$,

and $(1+x)^{-n}$ is the generating function of the sequence $\left\{ \binom{-n}{0}, \binom{-n}{1}, \binom{-n}{2}, \cdots \right\}$, where $\binom{-n}{r} = (-1)^r \binom{n+r-1}{r}$

Infact for any $n \in \mathbb{R}$, one can use Maclaurin series to get that

$$(1+x)^n = 1 + nx + n(n-1)\frac{x^2}{\lfloor 2} + n(n-1)(n-2)\frac{x^3}{\lfloor 3} + \cdots$$

$$= 1 + \sum_{r=1}^{\infty} \frac{n(n-1)(n-2)\cdots(n-r+1)}{\lfloor r} x^r$$

That is, $(1+x)^n$ is the generating function of the sequence $\left\{ \binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \cdots \right\}$, where $\binom{n}{r} = \frac{n(n-1)(n-2)\cdots(n-r+1)}{\lfloor r}$

★ If how many ways can we select, with repetitions allowed, s objects from n distinct objects?

For each n distinct objects, the geometric series $1+x+x^2+x^3+\cdots$ represents the possible choices (by the power of x, the number of choices). Considering all of the n distinct objects, the generating function is

$f(x) = (1+x+x^2+\cdots)^n$, and the coefficient of x^s in $f(x)$ gives the answer. Further,

$$f(x) = \left(\frac{1}{1-x}\right)^n$$

$$= (1-x)^{-n}$$

$$= 1 + \sum_{r=1}^{\infty} \binom{-n}{r} (-x)^r$$

$$= 1 + \sum_{r=1}^{\infty} (-1)^r \binom{n+r-1}{r} (-1)^r x^r$$

$$= \sum_{r=0}^{\infty} \binom{n+r-1}{r} x^r$$

so the coefficient of x^s is $\binom{n+s-1}{s}$

which we have already seen before.

★ Some useful applications of generating functions:-

i) Counting the number of compositions of a positive integer n. [Answer: 2^{n-1}]

ii) The number of composition of n, when n is a positive integer and each composition is a palindrome is given by $2^{\lfloor n/2 \rfloor}$

iii) Prove that $\binom{2n}{n} = \sum_{i=0}^{n} \binom{n}{i}$, when $n \in \mathbb{Z}^+$.

★ Exercises:-

1. Find the number of integer solutions for the following equations:

(a) $c_1 + c_2 + c_3 + c_4 = 20$, $0 \le c_i \le 7$ for all $1 \le i \le 4$

(b) $c_1 + c_2 + c_3 + c_4 = 20$, $0 \le c_i$ for all $1 \le i \le 4$, with c_2 and c_3 even

(c) $c_1 + c_2 + c_3 + c_4 + c_5 = 30$, $2 \le c_1 \le 4$ and $3 \le c_i \le 8$, for all $2 \le i \le 5$

(d) $c_1 + c_2 + c_3 + c_4 + c_5 = 30$, $0 \le c_i$ for all $1 \le i \le 5$, with c_2 even and c_3 odd.

2. Find the generating functions for the following sequences. (a) $\binom{8}{0}, \binom{8}{1}, \binom{8}{2}, \ldots, \binom{8}{8}$

(b) $\binom{8}{1}, 2\binom{8}{2}, 3\binom{8}{3}, \ldots, 8\binom{8}{8}$

(c) $1, -1, 1, -1, 1, -1, \ldots$

(d) $0, 0, 0, 6, -6, 6, -6, 6, \ldots$

(e) $1, 0, 1, 0, 1, 0, 1, \ldots$

(f) $0, 0, 1, a, a^2, a^3, \ldots, \quad a \neq 0$

3. Determine the sequence generated by the following generating functions.

(a) $f(x) = (2x-3)^3$ (b) $f(x) = \dfrac{x^4}{1-x}$

(c) $f(x) = \dfrac{x^3}{1-x^2}$ (d) $f(x) = \dfrac{1}{1+3x}$

(e) $f(x) = \dfrac{1}{3-x}$ (f) $f(x) = \dfrac{1}{1-x} + 3x^7 - 11$

4. In each of the following, the function $f(x)$ is the generating function for the sequence $a_0, a_1, a_2, \ldots$, whereas the sequence $b_0, b_1, b_2, \ldots$ is generated by the function $g(x)$. Express $g(x)$ in terms of $f(x)$.

(a) $b_3 = 3, \quad b_n = a_n, \; n \in \mathbb{N}, \; n \neq 3$

(b) $b_3 = 3, \; b_7 = 7, \; a_n = b_n, \; n \in \mathbb{N}, \; n \neq 3, 7$

(c) $b_1 = 1, \; b_3 = 3, \; b_7 = 7, \; b_n = 2a_n + 5,$
$n \in \mathbb{N}, \; n \neq 1, 3, 7$

(d) $b_1 = 1, \; b_3 = 3, \; b_n = 2a_n, \; n \in \mathbb{N}, \; n \neq 1, 3$

5. Determine the constant in $\left(3x^2 - \frac{2}{x}\right)^{15}$

6. Find the coefficient of x^{50} in $(x^7 + x^8 + x^9 + \ldots)^6$

7. Find the coefficient of x^{20} in $(x^2 + x^3 + x^4 + x^5 + x^6)^5$

8. Find the coefficient of x^{15} in each of the following.

(a) $x^3(1-2x)^{10}$ (b) $(x^3 - 5x)/(1-x)^3$ (c) $(1+x)^4/(1-x)^4$

9. In how many ways can 3000 identical envelopes be divided, in packages of 25, among four student groups so that each group gets at least 150, but not more than 1000, of the envelopes?

10. If a fair die is rolled 12 times what is the probability that the sum of the rolls is 30?

11. In how many ways can Traci select n marbles from a large supply of blue, red and yellow marbles (all of the same size) if the selection must include an even number of blue ones?

12. Verify that $(1-x-x^2-x^3-x^4-x^6)^{-1}$ is the generating function for the number of ways the sum n, $n \in \mathbb{N}$, can be obtained when single die is rolled an arbitrary number of times.

13. Show that $(1-4x)^{-\frac{1}{2}}$ generates the sequence $\binom{2n}{n}$, $n \in \mathbb{N}$

14. Let n be a (fixed) positive integer, with $n \geq 2$. If $1 \leq t \leq \lfloor n/2 \rfloor$, how many palindromes of n start with t?

15. Let $n \in \mathbb{Z}^+$, n odd. Can a palindrome of n have an even number of summands?

16. Let $n \in \mathbb{Z}^+$, n even. How many palindromes of n have an even number of summands? How many have an odd number of summands?

17. Shay rolls a fair die until she gets a 6. If random variable y counts the number of times Shay rolls the die until she gets her first 6, determine (a) the probability distribution for y; (b) $E(y)$; (c) σ_y.

18. Find a formula for the convolution of each of the following pairs of sequences.

(a) $a_n = 1$, $0 \le n \le 4$, $a_n = 0$, for all $n \geq 5$;

$b_n = n$, for all $n \in \mathbb{N}$

(b) $a_n = (-1)^n$, $b_n = (-1)^n$, for all $n \in \mathbb{N}$

[If $f(x) = \sum_{i=0}^{\infty} a_i x^i$ and $g(x) = \sum_{i=0}^{\infty} b_i x^i$, and $h(x) = f(x)g(x)$, then $h(x) = \sum_{i=0}^{\infty} c_i x^i$, where for all $k \geq 0$, one can write

$$c_k = \sum_{i=0}^{k} a_i b_{k-i}$$

The sequence $\{c_i\}_{i \geq 0}$ is called the convolution of the sequences $\{a_i\}_{i \geq 0}$ and $\{b_i\}_{i \geq 0}$.]